VIGINT ENTERPRISE COGNITIVE SURVEILLANCE PLATFORM

Intelligent Compliance & Continuous Rule-Based Monitoring System

Architectural Blueprint, Cognitive AI Stack (RelateAnything & Laya/JEV), Live RTSP Stream Decoupling, and Complete Implementation Plan for Institutional & Campus Security

Target Domain: Campus, Institutional & High-Security Zones	**Decision Latency:** Sub-35ms Non-Autoregressive
Core AI Engine: RelateAnything (RAM) + Laya / JEV	**Feed Mode:** Manually Ingested RTSP Streams (Decoupled)
Evidence Standard: Section 65B IEA / BNSS 2023 with SHA-256	**Operational Page:** /compliance Rule Studio & SOC Stream

1. Executive Overview & Problem Formulation

Modern institutional environments—such as university campuses, medical centers, server complexes, and government research laboratories—deploy dozens of IP and RTSP cameras. However, the vast majority of these cameras function as passive digital recorders. Operators only review footage post-incident, rendering physical security reactive rather than preventive. When standard Video Management Systems (VMS) attempt automated alerts, they rely almost exclusively on crude motion detection or basic bounding box classifiers. These legacy approaches suffer from two fatal weaknesses: an overwhelming rate of false positives (triggered by shadows, animals, foliage, or normal authorized movement) and an utter inability to understand **relational context** (such as distinguishing between an individual merely standing next to a perimeter fence versus actively scaling it).

To solve this, we introduce the **VIGINT Compliance & Vulnerability System**. This system transforms passive video feeds into an active, rule-enforcing cognitive network. Institutional administrators can configure exact spatial boundaries, temporal schedules, and complex relational rules. A dedicated multi-level AI cognitive stack continuously scans live feeds, detects non-compliant actions, and immediately delivers flagged alerts to the central operations center. Each alert is enriched with a high-resolution highlighted keyframe and an automatically stitched, cryptographically sealed video clip capturing the incident alongside its preceding and succeeding seconds.

Core Scope Directive: Rather than merging all 30 statewide traffic feeds into this intensive analytical pipeline, the compliance system operates exclusively over **manually registered RTSP camera feeds** (e.g., campus gates, lab entrances, server rooms, perimeter walls). This guarantees low compute overhead, ultra-low latency, and pinpoint operational focus.

2. Deep-Dive AI Cognitive Stack & Model Architecture

The cognitive surveillance engine operates as a three-tier hierarchical pipeline designed to extract visual primitives, build semantic relationship graphs, and execute deterministic, sub-35ms rule evaluations.

Tier 1: Visual Primitive Perception (YOLOv8/v11 + YOLO-Pose + ByteTrack)

Before evaluating rules, the system extracts foundational spatial and kinematic entities. Object detection is handled by a customized YOLOv8/v11 backbone fine-tuned for institutional environments, detecting persons, vehicles, personal protective equipment (safety helmets, high-visibility vests), backpacks, suitcases, weapons, and hazard indicators (fire and smoke). Concurrently, **YOLOv8-Pose** identifies 17 anatomical keypoints per individual, which is vital for detecting abnormal physical events such as climbing over turnstiles or boundary fences (evaluated by ankle/knee coordinates relative to horizontal barriers), slip-and-fall injuries (torso axis aligned parallel to the ground plane for over 3 seconds), or physical altercations. The **ByteTrack** multi-object tracking algorithm maintains consistent track IDs across frames, computing real-time dwell times, velocity vectors, and directional trajectories.

Tier 2: Relational Scene Graphs via RelateAnything Model (RAM)

Isolated bounding boxes cannot convey human behavior. The **RelateAnything Model (RAM)** bridges this gap by extracting visual relationship triplets in the form of <Subject, Predicate, Object>. RAM evaluates spatial proximity and bounding box Intersection-over-Union (IoU) between all detected entity pairs. Candidate pairs are evaluated against a rich semantic prior matrix categorized into four distinct behavioral classes: (1) *Spatial/Topological* (e.g., <person, inside_zone, server_rack>), (2) *Possession & Asset Handling* (e.g., <person, carrying, equipment>, <bag, abandoned_near, entrance>), (3) *EHS & PPE Compliance* (e.g., <worker, without, helmet>), and (4) *Security Violations* (e.g., <person, climbing, perimeter_fence>, <individual, forcing_open, exit_door>). This converts unstructured pixel streams into a structured, lightweight Scene Graph for every analyzed frame.

Tier 3: Smart Decision Making via Laya & JEV System-1 Architecture

A critical design choice is avoiding autoregressive Vision-Language Models (such as GPT-4V, LLaVA, or Claude Vision) in the continuous live inference loop. Generative VLMs incur extreme latency (800ms–2500ms per frame), require massive GPU clusters, and suffer from unpredictable hallucinations. Instead, VIGINT deploys **Laya and JEV (Joint Embedding Vision & Non-Autoregressive Decision Engine)**. Laya operates a dual-stream System-1 reasoning pipeline that evaluates candidate scene graphs against rule conditions in a **single forward pass**, achieving latency under **18.4ms**. Outputs are strictly typed JSON records specifying violation state, calibrated confidence score (via Reinforcement Learning from Calibrated Decisions - RLCD), and mandatory Standard Operating Procedure (SOP) dispatch instructions. Edge-case false positives are actively suppressed by Laya's strictly proper scoring calibration.

Metric / Feature	Autoregressive VLMs (LLaVA / GPT-4V)	VIGINT Stack (RelateAnything + Laya/JEV)
Inference Latency	800 ms – 2,500 ms (Unusable for live RTSP)	**18 ms – 35 ms** (Real-time continuous)
Output Structure	Unstructured text with hallucination risk	**Deterministic typed JSON** with strict schema
Compute Footprint	High-end enterprise GPUs (40GB–80GB VRAM)	**Mid-tier GPU / AVX-512 CPU** (4GB–8GB VRAM)
Confidence Calibration	Poor / uncalibrated generative certainty	**RLCD Calibrated Scoring** (Suppresses false alarms)

3. Rule-Based Compliance Engine Specification

The compliance engine evaluates live streams against declarative, user-defined rule matrices. Rules are organized across five core institutional operational domains:

Compliance Category	Target Violations	Algorithmic Detection Mechanics
1. Zone Intrusion & Perimeter Breach	Unauthorized presence in server rooms, scaling perimeter walls, entry into electrical vaults.	Polygonal geofencing (Ray-Casting) + YOLO-Pose (elevated limb coordinates) + RelateAnything.
2. Occupational Safety & PPE	Personnel lacking hard hats or safety vests in designated engineering/lab zones.	YOLOv8 PPE detector + RelatePredicate <worker, without, helmet>.
3. Loitering & Temporal Curfews	Individuals remaining in parking lots > 5 mins, or inside classrooms past midnight.	ByteTrack persistent dwell timer coupled with system schedule calendar validation.
4. Asset Security & Abandoned Items	Unattended luggage (> 60s without associated person within 2m), unauthorized device removal.	RelateAnything spatial distance tracking: <bag, abandoned_by, person>.
5. Access Control & Anti-Tailgating	Piggybacking through turnstiles or secure RFID doors on a single badge-in.	Spatial proximity tracking of two human centroids crossing a tripwire in < 3.0 seconds.

4. Decoupled Live Streaming & Evidence Packaging Pipeline

Continuous neural processing at 30 FPS across multiple high-definition RTSP streams causes memory saturation and thermal throttling. To deliver 24/7 reliability, VIGINT employs a **Decoupled Asynchronous Streaming Pipeline**:

1. High-Speed Circular Ring Buffer (Ingestion Thread - 30 FPS): An isolated background thread ingests raw H.264/H.265 RTSP streams over TCP, continuously populating an in-memory circular ring buffer holding the most recent 300 frames (10 seconds of

full-framerate video). Older frames are overwritten continuously with zero disk I/O.

2. Downsampled Inference Cadence (3 to 5 FPS): An asynchronous consumer thread samples every 6th frame from the ring buffer, downsamples it to 640x640, and submits it to the YOLO $\rightarrow$ RAM $\rightarrow$ Laya/JEV pipeline. This achieves an 80%–85% reduction in compute demand while maintaining a rapid detection latency under 250 milliseconds.

3. Automated Evidence Assembly & Cryptographic Sealing: The moment a rule violation is flagged by Laya/JEV, the system instantly triggers two automated packaging processes: (a) *Annotated Keyframe*: The violating frame is stamped with bounding boxes, relational interaction vectors, timestamp, and violated rule ID, saved as high-resolution JPEG; (b) *Synchronous Evidence Clip*: The engine slices 150 frames before the event (5s pre-incident context) and 150 frames after the event (5s post-incident resolution) from the ring buffer, concatenates them, and encodes an MP4 video clip. A cryptographic SHA-256 hash is computed immediately and stored in the database, guaranteeing statutory admissibility under Section 65B of the Indian Evidence Act and Section 63 of the Bharatiya Sakshya Adhiniyam (BNSS 2023).

5. User Interface & Operations Center Architecture (/compliance)

The new dedicated /compliance frontend module provides institutional administrators and security operators with a comprehensive, unified operational command surface divided into three synchronized workspaces:

A. Interactive Rule Studio: Allows administrators to create, test, and activate compliance rules without writing code. Features an interactive HTML5 canvas overlay where users can select any registered manual RTSP camera feed, capture a live reference snapshot, and draw custom polygonal zones (e.g., Restricted Server Room Perimeter) or directional tripwires. Rule conditions are configured via intuitive controls: actor selection (person, vehicle, worker), relational interaction predicates, schedule calendars (operating hours vs. curfew), and dwell time thresholds.

B. Live Compliance Monitoring Wall: Renders the active institutional RTSP streams in a responsive grid. Monitored zones are dynamically rendered with soft green translucent overlays during compliant conditions. When an unauthorized individual breaches a boundary or violates a rule, the affected polygon turns pulsating emergency red, accompanied by a real-time relational vector arrow pointing from the subject to the affected asset.

C. Incident Response & Evidence Center: A low-latency live incident stream connected via WebSockets. When a violation occurs, an audible alarm triggers at the central console. The operator is presented with the annotated still keyframe, the extracted relational triplet text (e.g., <person_14 scaling perimeter_wall>), and an integrated HTML5 video player streaming the 10-second buffered evidence clip. From this console, operators can acknowledge the alert, dispatch security personnel with one click, or export an official statutory forensic report with verified hashes.

6. Database Architecture & API Specification

The system is powered by two primary database entities designed for high query performance and forensic integrity:

```
-- 1. Compliance Rules Table
CREATE TABLE compliance_rules (
    rule_id TEXT PRIMARY KEY, name TEXT NOT NULL, category TEXT NOT NULL,
    severity TEXT NOT NULL, enabled BOOLEAN DEFAULT 1, camera_ids TEXT NOT NULL,
    spatial_config TEXT NOT NULL, temporal_config TEXT NOT NULL,
    relational_config TEXT NOT NULL, created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 2. Compliance Violations & Evidence Table
CREATE TABLE compliance_violations (
    violation_id TEXT PRIMARY KEY, rule_id TEXT NOT NULL, camera_id TEXT NOT NULL,
    camera_name TEXT NOT NULL, severity TEXT NOT NULL, timestamp REAL NOT NULL,
    relational_triplet TEXT, calibrated_confidence REAL NOT NULL,
    keyframe_url TEXT NOT NULL, footage_clip_url TEXT NOT NULL, sha256_hash TEXT NOT NULL,
    status TEXT DEFAULT 'OPEN', assigned_officer TEXT, resolution_notes TEXT,
    FOREIGN KEY(rule_id) REFERENCES compliance_rules(rule_id)
);
```

REST & WebSocket Interface Endpoints:

- GET /api/compliance/rules — Retrieve all active and disabled institutional compliance rules.
- POST /api/compliance/rules — Create or update a rule with polygonal geofences and temporal schedules.
- GET /api/compliance/violations — Query historical violations filterable by date, camera, severity, and status.
- POST /api/compliance/violations/{id}/action — Execute triage actions (Acknowledge, Dispatch, or Resolve).
- WS /ws/compliance — High-speed bi-directional WebSocket push feed delivering immediate incident notifications.

7. Hardware Sizing & Compute Budget Feasibility

Hardware Configuration	Concurrent RTSP Feeds	Pipeline Throughput & Latency	Operational Recommendation
Mid-Tier Consumer GPU (NVIDIA RTX 4060 / 3060 8GB)	4 to 6 RTSP streams simultaneously	4 FPS cadence per stream Total pipeline latency: ~38ms VRAM load: ~3.8 GB	**Highly Cost-Effective** Ideal for departmental campus buildings, server rooms, and gates.
Enterprise / Workstation GPU (NVIDIA RTX 4090 / A5000 24GB)	12 to 16 RTSP streams simultaneously	5 FPS cadence per stream Total pipeline latency: ~22ms VRAM load: ~7.2 GB	**Maximum Scale** Recommended for centralized institutional Security Operations Centers.
CPU Only (AVX2 / AVX-512) (Intel Core i7/i9 or AMD Ryzen 9)	1 to 2 RTSP streams simultaneously	2–3 FPS cadence with OpenVINO / ONNX Total latency: ~110ms System RAM: ~4.5 GB	**Edge / Pilot Deployments** Viable for single entrance gates or standalone laboratory units.

8. Four-Phase Step-by-Step Implementation Roadmap

To execute this blueprint without any friction or architectural disruption, implementation will proceed in four phases:

Phase	Workstream	Key Technical Deliverables & Milestones
Phase 1	Core Reasoning & Engine (Backend)	• Define ComplianceRule and ComplianceViolation SQLite tables. • Build compliance_engine.py integrating polygonal geofencing with relate_anything_engine.py. • Implement the in-memory circular ring buffer (300 frames) for manual RTSP streams.
Phase 2	REST & WebSocket API Layer (Backend)	• Develop routers/compliance.py with CRUD endpoints for rules and historical violation logs. • Implement the WebSocket alert broadcaster (/ws/compliance) for real-time dispatch. • Implement automated 10-second MP4 evidence slicing with cryptographic SHA-256 calculation.
Phase 3	Compliance UI Studio & SOC Console (Frontend)	• Create the new Next.js route at frontend/app/compliance/page.tsx. • Build the polygonal geofencing drawing canvas for rule creation over frozen camera snapshots. • Construct the real-time Incident Drawer with keyframe inspection, HTML5 buffer playback, and SOP dispatch.
Phase 4	Field Calibration & Validation	• Connect active campus/institutional RTSP streams to verify multi-stream stability. • Calibrate confidence thresholds for loitering, intrusion, and PPE compliance. • Validate end-to-end alert latency (sub-500ms from physical event to SOC dashboard presentation).

Document Sign-off: *VIGINT Advanced Engineering & Computer Vision Research Group*
Status: *Approved Technical Specification & Research Architecture | Ready for Phase 1 Execution*